

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions

Remote

- Built a resilient **Puppeteer/Node.js workflow engine** for MCA company-registration filings, reducing manual filing time by **85%** through state recovery, retries, and public-site API lookups.
- Optimized an **AI-assisted company-registration pipeline** for unique company-name suggestions, descriptions, and filing-field drafts, reducing manual drafting/rework across customer workflow preparation.
- Containerized services with Docker, deployed to Google Cloud Run, and automated releases through GitHub Actions, improving deployment velocity by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built MRI segmentation tooling for **95 anatomical regions**, integrating model inference, preprocessing, generated masks, and 3D visualization into an existing application.
- Reduced processing runtime by **87%** using asynchronous and parallel execution across expensive image-processing workflows.
- Debugged integration issues across PyTorch/ONNX-style inference, label mapping, VTK/wxPython UI paths, and large-volume medical data handling.

Open Source

Contributions 

- Contributed maintainer-reviewed patches to **Invesalious** medical-imaging UI/compatibility paths, **Tiled** editor workflows, and **Kuadrant/Kyverno** cloud-native integration code, plus parsing/security fixes such as Zip Slip path validation.

Projects

Seaweed: Contest Platform | *TypeScript, React, Next.js, Go, PostgreSQL, Judge0, Firebase*

GitHub 

- Built a **React/Next.js assessment UI** with auth-gated problem pages, code editor, submissions, admin review/shortlisting, and live ranked leaderboards.
- Implemented Go judging APIs with Judge0, PostgreSQL rankings/materialized views, Redis-backed runtime state, and Docker/Kubernetes deployment for **500+ concurrent users** and **sub-5s P95 verdict latency**.

Hyoka | *Python, FastAPI, SQLAlchemy, Alembic, PostgreSQL, Next.js, TypeScript, Docker*

- Built a **reliability control plane for AI agents** with trace/event ingestion, authenticated APIs, suites, candidates, validation runs, gate decisions, promotions, audit logs, and a Next.js operations dashboard.
- Implemented SQLAlchemy/Alembic metadata models, project-scoped API keys, content-addressed artifacts, signed manifests, worker-owned execution leases, and Docker Compose deployment with Postgres-backed services.

Postificus | *Go, Echo, Rod, PostgreSQL, Redis, RabbitMQ, Docker, Prometheus*

GitHub 

- Built Go/Echo services for a content distribution platform using browser automation, REST APIs, RabbitMQ workers, Redis autosave, and PostgreSQL persistence.
- Added retries, DLQs, circuit-breaker isolation, health checks, and Prometheus metrics so unreliable external browser/API workflows stayed recoverable across distributed workers.

Technical Skills

Languages: Python, TypeScript, Go, SQL, C++

Frameworks: React, Next.js, Node.js, FastAPI, Echo, Gin, PyTorch

Tools: PostgreSQL, SQLAlchemy, Redis, RabbitMQ, Docker, GCP, AWS, GitHub Actions, Firebase, Prometheus